and elongated. The nuclear membrane and the nucleolus reappear and thus two daughter nuclei are formed.

e. Cytokinesis I: The cytoplasmic division occurs and two haploid cells are formed.

B. Homotypic Division

In this division, the two haploid cells formed during first meiotic division divide into four haploid cells. The daughter cells are similar to parent cell in the chromosome number (Homotypic). It consists of five stages.

a. Prophase – II b. Metaphase – II
c. Anaphase – II d. Telophase – II
e. Cytokinesis – II

a. Prophase II: The centriole divides into two, each one moves to opposite poles. Asters and spindle fibres appear. Nuclear membrane and nucleolus disappear.

b. Metaphase II: The chromosomes get arranged on the equator. Two chromatids are separated.

c. Anaphase II: The separated chromatids become daughter chromosomes and move to opposite poles

d. Telophase II: The daughter chromosomes are centered. The nuclear membrane and the nucleolus appear.

e. Cytokinesis II: Two cells are formed from each haploid daughter cell, resulting in the formation of four cells with haploid number of chromosomes.

Significance of Meiosis

- The constant number of chromosomes in a given species is maintained by meiotic division.
- Genetic valiation is produced due to crossing over within the species which is transmitted from one generation to next generation.

Table 18.3 Differences between Mitosis and Meiosis

Mitosis	Meiosis
Occurs in somatic cells.	Occurs in reproductive cells.
Involved in growth and occurs continuously throughout life.	Involved in gamete formation only during the reproductively active age.
Consists of single division.	Consists of two divisions.
Two diploid daughter cells are formed.	Four haploid daughter cells are formed.
The chromosome number in the daughter cell is similar to the parent cell (2n).	The chromosome number in the daughter cell is just half (n) of the parent cell.
Identical daughter cells are formed.	Daughter cells are not similar to the parent cell and are randomly assorted.

Points to Remember

- In general plant tissues are classified into two types namely Meristems or Meristematic tissues and Permanent tissues.
- Permanent tissues are of two types: simple tissue and complex tissue.
- Simple tissues are homogeneous-composed of structurally and functionally similar cells. eg., Parenchyma, Collenchyma and Sclerenchyma.
- Complex tissues are made of more than one type of cells that work together as a unit. They are Xylem and Phloem
- Animal tissues can be grouped into four basic types on the basis of their structure and functions. Epithelial tissue, Connective tissue, Muscular tissue and Nervous tissue.
- Simple epithelium formed of single layer of cells is divided into following types. They are squamous epithelium cuboidal epithelium, columnar epithelium, ciliated epithelium and glandular epithelium
- Compound Epithelium consists of more than one layer of cells and gives a stratified appearance.
- The three types of cell division that occur in animal cells are Amitosis, Mitosis and Meiosis.

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